THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 11

# Probability, Universality, and Levels of Description

Note Template

Self-Study Curriculum

**How to use this template**

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

## 1. Why "I already use probability" is not a reason to skip it

*state the using-vs-having distinction in your own words*

## 2. The three interpretations of probability

Frequency:

Degree of belief (Bayesian):

Measure (Kolmogorov):

What is settled: ______________ What is contested: ______________

## 3. Law of Large Numbers vs. Central Limit Theorem

LLN says:
CLT says (and why it is stranger):

## 4. Concentration of measure, why averaging works

Averaging works because:
Averaging fails when: (two conditions)

## 5. Universality

*define it in one sentence, without the word "universality"*
The mechanism (coarse-graining / renormalization):
The connection to the Chapter 10 fixed point:

## 6. Why is biology modelable at all despite the molecular mess?

*answer in terms of coarse-graining*

## 7. Levels of description

Why can I do population biology without quantum mechanics?

## 8. What universality does NOT explain (what stays open)